Forensic analysis report

This is an analysis of what could be shown from the file. It is NOT a certificate of authenticity.

File	D_roundtrip_amount_altered.pdf
SHA-256	`0c37c281f95a76aa38879178185361205c70df7a4d46bfaf3f97478bec02b20b`
Analysed	2026-11-06T20:00:00Z by Doc Forensics Engine 0.1.0 (ruleset 2026.11-r1)

REPORT CATEGORY

Confirmed change

A change is demonstrated by a trusted original or by cryptographic evidence.

Why this category

* 1 text difference from the original - Relative to the supplied original, these passages were added, removed or changed in the examined document. The original and the examined text are shown side by side.
* Amount differs from the original - A amount value in the examined document differs from the corresponding value in the supplied original.
* 1 further indicator(s) are also listed for review.

No probability or confidence score is given: none of these checks has been statistically calibrated.

1. DOCUMENT OVERVIEW

Size	36611 bytes
SHA-512	`e7e8bbaf63d39545adfde492e797411d1d8f9c925e39f2c933f6fb52be10758e2323f0f5aa6c07a0cacc8affc746d76bc1475c84cb4e1e8b85953c1754cc213b`
SHA-1	`196c2bad8e68034dc17aa830f75eee73f7a53fee`
MD5	`b061e4f6e1fe09d47a5b84441ee216b9`
PDF version	1.7
Pages	1
Revisions	1 (no incremental updates)
Creator	Writer
Producer	LibreOffice 24.2
Created	2026-11-06 18:33:43 UTC

2. CREATION-TOOL AND METADATA EVIDENCE

Labels are self-asserted and can be edited or forged. Use of a tool is not evidence of falsification.

[INFORMATION] Software labels found in metadata

Detected:

Info /Creator: "Writer" -> LibreOffice / OpenOffice Writer (Word processor / typesetting); Info /Producer: "LibreOffice 24.2" -> LibreOffice / OpenOffice Writer (Word processor / typesetting)

Establishes:

The file carries self-asserted labels that are consistent with the software named above having created or processed it.

Unknown:

Labels can be edited, stripped or forged, and they do not show what was done to the content. Use of any of these tools is not evidence that the document was falsified.

3. SIGNATURE INFORMATION

No digital signatures were found, so there is no cryptographic evidence to check integrity against.

A valid signature does not show that the signed information is true: a document made with false information and then signed still verifies.

4. EDITING AND REVISION INDICATORS

Indicators are prompts for human review, not proof of manipulation.

[INFORMATION] Single revision: no incremental-update history

Detected:
The file contains exactly one revision.

Establishes:
No earlier version of this document can be recovered from the file itself.

Unknown:
Absence of history does not mean the document was never edited: editors that rewrite the whole file (re-saving, converting, printing to PDF) leave no incremental history. A PDF that was regenerated cannot be distinguished from an original on this evidence alone.

[INFORMATION] Unreferenced objects remain in the file

Detected:
17x Tagged-PDF structure element (obj 6, 9, 12, 15, 18, 21...)

Establishes:
These objects are live in the cross-reference table but nothing in the document refers to them, so no viewer displays them.

Unknown:
Writers that do not garbage-collect leave unreferenced objects behind routinely; deleted content is one cause among several. See the recovered text preview, if any.

[INFORMATION] 2 font face(s) used

Detected:
BAAAAA+LiberationSans-Bold - TrueType - embedded - subset BAAAAA - pages 1 | CAAAAA+LiberationSans - TrueType - embedded - subset CAAAAA - pages 1

Establishes:
These are the font resources the pages declare.

Unknown:
Fonts show how text was typeset, not what it says or who wrote it. Software substitutes fonts during conversion, so the same visible document can legitimately use different fonts in different files.

[INFORMATION] Page composition

Detected:
1 page(s): vector text.

Establishes:
Shows whether each page is typeset text, a scanned picture, or a picture with text on it.

Unknown:
Whether a scan faithfully shows its source. Image-only pages cannot be text-compared without OCR, which this tool does not perform.

5. FILE COMPARISON (ORIGINAL VERSUS SUBMITTED)

Reference	A_original.pdf (PDF); supplied by the examiner and unverified by this tool
File bytes	different from the reference
Document meaning	differences found

Side-by-side differences

* page 1: 4,250.00 -> 42,500.00

[ESTABLISHED] 1 text difference from the original

Detected:
"4,250.00" -> "42,500.00" [amount] (page 1)

Establishes:
Relative to the supplied original, these passages were added, removed or changed in the examined document. The original and the examined text are shown side by side.

Unknown:
Whether the difference is a falsification, a legitimate later version, or an artefact of conversion. Conversion

between formats can re-flow text, and name detection is pattern-based. This tool cannot verify that the reference you supplied is itself authentic; the comparison is only as reliable as that reference.

2 highlighted region(s) require review (section 7).

[ESTABLISHED] Amount differs from the original

Detected:

total received: "ZMW 4,250.00" -> "42,500.00" (page 1)

Establishes:

A amount value in the examined document differs from the corresponding value in the supplied original.

Unknown:

Whether the difference is deceptive or a legitimate update. This tool cannot verify that the reference you supplied is itself authentic; the comparison is only as reliable as that reference.

[INDICATOR] 13 region(s) of the rendered pages differ from the original

Detected:

page 1: 13 region(s), 4.9% of pixels changed

Establishes:

Parts of the rendered pages look different from the original. Where the text layer is identical or unavailable, these regions are the only place a change could be visible.

Unknown:

Rendered comparison cannot tell edits from re-rendering or re-scanning differences, so these regions are for human review (use the side-by-side view); they are not proof. No similarity score is computed.

26 highlighted region(s) require review (section 7).

[INFORMATION] Embedded pictures match the original

Detected:

obj 4 ? obj 5 (160x200, page 1): same picture, different encoding

Establishes:

The pictures in the examined document are the same pictures as in the original (possibly re-encoded).

Unknown:

Small local retouching below the comparison's sensitivity is not excluded.

6. IMAGES AND SCANS

Image indicators are never proof of editing; each method has documented blind spots.

[INFORMATION] 1 embedded JPEG image(s) inspected

Detected:

obj 5 (page 1): 160x200, 4:2:0, progressive, standard tables ~ quality 90

Establishes:

How each JPEG was written: compression tables, sampling and metadata.

Unknown:

Tables show the LAST compression only. Non-standard tables are normal for cameras, phones and some scanners.

7. HIGHLIGHTED REGIONS REQUIRING REVIEW

* page 1 (submitted): x 468, y 357, 53 x 12 pt - examined: "42,500.00"
* page 1 (reference): x 472, y 314, 47 x 12 pt - original: "4,250.00"
* page 1 (submitted): x 438, y 36, 107 x 134 pt - visual difference
* page 1 (reference): x 438, y 36, 107 x 134 pt - visual difference
* page 1 (submitted): x 71, y 71, 205 x 62 pt - visual difference
* page 1 (reference): x 71, y 71, 205 x 62 pt - visual difference
* page 1 (submitted): x 71, y 143, 259 x 107 pt - visual difference
* page 1 (reference): x 71, y 143, 259 x 107 pt - visual difference
* page 1 (submitted): x 357, y 241, 45 x 36 pt - visual difference
* page 1 (reference): x 357, y 241, 45 x 36 pt - visual difference
* page 1 (submitted): x 438, y 241, 45 x 9 pt - visual difference
* page 1 (reference): x 438, y 241, 45 x 9 pt - visual difference
* page 1 (submitted): x 71, y 259, 80 x 18 pt - visual difference

* page 1 (reference): x 71, y 259, 80 x 18 pt - visual difference
* page 1 (submitted): x 438, y 259, 80 x 18 pt - visual difference
* page 1 (reference): x 438, y 259, 80 x 18 pt - visual difference
* page 1 (submitted): x 71, y 286, 107 x 62 pt - visual difference
* page 1 (reference): x 71, y 286, 107 x 62 pt - visual difference
* page 1 (submitted): x 357, y 286, 9 x 9 pt - visual difference
* page 1 (reference): x 357, y 286, 9 x 9 pt - visual difference
* page 1 (submitted): x 375, y 286, 18 x 9 pt - visual difference
* page 1 (reference): x 375, y 286, 18 x 9 pt - visual difference
* page 1 (submitted): x 438, y 286, 107 x 9 pt - visual difference
* page 1 (reference): x 438, y 286, 107 x 9 pt - visual difference
* page 1 (submitted): x 375, y 303, 18 x 18 pt - visual difference
* page 1 (reference): x 375, y 303, 18 x 18 pt - visual difference
* page 1 (submitted): x 438, y 303, 98 x 45 pt - visual difference
* page 1 (reference): x 438, y 303, 98 x 45 pt - visual difference

8. LIMITATIONS

* This tool reports what could be demonstrated from the supplied bytes. It cannot certify a document as genuine.
* Metadata, tool labels, XMP history and dates are self-asserted by the file and can be edited or forged.
* Without a trusted original, the document's original contents are unknown; indicators compare the file only against itself.
* A PDF that was fully regenerated (printed to PDF, converted, optimised) carries no recoverable history of earlier versions. This is a limit of the evidence, not a sign of innocence or guilt.
* A valid digital signature shows the signed bytes are unchanged since signing; it does not show that what was signed is true.
* Indicators are prompts for human review and are not accuracy-rated: none of the checks has been statistically calibrated, so no probability or confidence percentage is given.

9. REPRODUCIBILITY

Tool	Doc Forensics Engine 0.1.0
Ruleset	2026.11-r1
PDF library	com.tom-roush:pdfbox-android:2.0.27.0
Crypto library	org.bouncycastle:bcprov-jdk18on:1.83
Trust set	test root CA (fixture generator)
Checks run	file fingerprints (SHA-256/512, SHA-1, MD5); PDF metadata and creation-tool labels; revision / incremental-update history; cross-reference integrity, page structure, fonts, unreferenced objects; text, fonts and paint order (covered, hidden, overlaid content); embedded images: JPEG structure, placement over scans, noise consistency; digital signatures (byte ranges, CMS verification, certificate trust, changes after signing); comparison with the supplied reference (file)

The complete findings are attached to this PDF as report.json (schema doc-forensics-report/1). The examined document itself is not included.